

# Math 161

## Midterm Exam 2

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1. Determine whether each of the following limits exist. If it does, what is its value? (Please justify your answer; it is possible to avoid  $\epsilon$ - $\delta$  arguments.)

(a)  $\lim_{x \rightarrow -2} \frac{x^2 - 3x - 10}{x + 2}$

(b)  $\lim_{x \rightarrow \infty} \frac{1 - x^2}{2x - 3}$

(c)  $\lim_{x \rightarrow \infty} \frac{x - 1}{x^2 + 1}$

2. Suppose that  $f : \mathbb{R} \rightarrow \mathbb{R}$  is continuous at 0, and that  $f(x + y) = f(x) + f(y)$  for all  $x, y \in \mathbb{R}$ . Prove that  $f$  is continuous at  $a$  for any  $a \in \mathbb{R}$ .

3. Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be a continuous function satisfying  $f(0) = 3$  and  $f(2) = 1$ . Prove that there exists  $x \in [0, 2]$  such that  $f(x) = x^2$ .